Loss function

$$e = (y_i - \hat{y}_i)^2$$

Cost function

$$f = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$\hat{y}_i = \hat{m} x_i + \hat{b}$$

So

$$f = \sum_{i=1}^{n} \left[y_i - (\hat{m} x_i + \hat{b}) \right]^2$$

In terms of mean square error (MSE)

$$f = \frac{1}{n} \sum_{i=1}^{n} \left[y_i - (\hat{m} x_i + \hat{b}) \right]^2 \quad \text{—(1)}$$

We need to find the values of $\hat{m}$ and $\hat{b}$ for which the cost function is the minimum. For this, we can formulate the following mathematical program for the unconstrained optimization problem

$$\underset{\hat{m}, \hat{b}}{\operatorname{argmin}} \; f = \frac{1}{n} \sum_{i=1}^{n} \left[y_i - (\hat{m} x_i + \hat{b}) \right]^2 \quad \text{—(2)}$$

This optimization problem can be solved using gradient descent algorithm.

The gradient descent steps are as follows:-

1- Let $x = \begin{bmatrix} \hat{m} \\ \hat{b} \end{bmatrix}$ and let the starting point b $x = \begin{bmatrix} m_0 \\ b_0 \end{bmatrix}$

2- Iteration

$$x_{i+1} = x_i - \lambda \nabla f \big|_{x_i}$$

3- If $|f(x_{i+1}) - f(x_i)| < G$ then stop

Else perform iteration 2

The gradient ∇f in step 2 can be determined as follows:-

$$\nabla f = \begin{bmatrix} \partial f / \partial m \\ \partial f / \partial b \end{bmatrix} \quad \text{——— (3)}$$

From (1)

$$\frac{\partial f}{\partial m} = \frac{2}{n} \sum_{i=1}^{n} \left[y_i - (\hat{m} x_i + \hat{b}) \right] (-x_i)$$

$$\frac{\partial f}{\partial m} = -\frac{2}{n} \sum_{i=1}^{n} x_i \left[y_i - (\hat{m} x_i + \hat{b}) \right] \quad \text{——— (4)}$$

Similarly,

$$\frac{\partial f}{\partial b} = \frac{2}{n} \sum_{i=1}^{n} \left[y_i - (\hat{m} x_i + \hat{b}) \right] (-1)$$

$$\frac{\partial f}{\partial b} = -\frac{2}{n} \sum_{i=1}^{n} \left[y_i - (\hat{m} x_i + \hat{b}) \right] \quad \text{——— (5)}$$

Using (4) and (5) in (3)

$$\nabla f = \begin{bmatrix} \partial f / \partial m \\ \partial f / \partial b \end{bmatrix} = \begin{bmatrix} -\frac{2}{n} \sum_{i=1}^{n} x_i \left[y_i - (\hat{m} x_i + \hat{b}) \right] \\ -\frac{2}{n} \sum_{i=1}^{n} \left[y_i - (\hat{m} x_i + \hat{b}) \right] \end{bmatrix}$$

$m_0 = 3.7426 \quad b_0 = 54.8732$

Learning rate $= \lambda = 0.01635$

		Iteration 1			Iteration 2		
x	y	$\hat{y}$	$y-\hat{y}$	$x[y-\hat{y}]$	$\hat{y}$	$y-\hat{y}$	$x[y-\hat{y}]$
2	69						
9	98						
5	82						
5	77						
3	71						
7	84						
1	55						
8	94						
6	84						
2	64						
		$\sum_{i=1}^{n}(y-\hat{y}) =$				Σ	

$$\text{grad } m = \frac{-2}{n} \times \sum_{i=1}^{n} x(y-\hat{y})$$

$$\text{grad } b = -\frac{2}{n} \times \sum_{i=1}^{n} (y-\hat{y})$$

$$m_1 = m_0 - \lambda \times \text{grad } m$$

$$b_1 = b_0 - \lambda \times \text{grad } b$$

$$\text{grad } m = -\frac{2}{n} \sum_{i=1}^{n} x(y-\hat{y})$$

$$\text{grad } b = -\frac{2}{n} \sum_{i=1}^{n} (y-\hat{y})$$

$$m_2 = m_1 - \lambda \times \text{grad } m_1$$

$$b_2 = b_1 - \lambda \times \text{grad } b$$